

Literal Native-Atom Path Archives for the TPC-15 Hard Packet

Local-to-Global Conservation, Metadata Intertwining, and First-Failure Certificates

Liang Wang

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Abstract

TPC-119 gave a sufficient canonical-matrix criterion for reconnecting the later TPC synthesis to the original fixed-shift hard packet, but the audited repository did not contain the complete matrix. We make the missing object local. Every declared partition, polarization, content split, frequency split, boundary rule and return rule is represented by an exact sparse stage matrix T_j , and the path-expanded archive records every nonzero edge rather than only its final aggregate. If

$$\mathbf{1}^T T_j = \mathbf{1}^T$$

at every stage and every edge intertwines the native key, prescribed shift and physical normalization, then the composite $M = T_s \cdots T_1$ is coefficientwise conservative and gives the exact TPC-119 reconnection identity. We prove the defect formula

$$\mathbf{1}_s^T M - \mathbf{1}_0^T = \sum_{j=1}^s (\mathbf{1}_j^T T_j - \mathbf{1}_{j-1}^T) T_{j-1} \cdots T_1,$$

which separates a global defect into typed local defects. Sharp finite examples show why equality on one coefficient vector, a conservative final composite, or an untagged scalar archive does not verify the local path statement. The theorem and exact rational audit are L0; identifying the required TPC fields is an L1 interface. The actual growing branch tables remain absent, so the literal archive is presently NOT-TESTABLE. No L2 fixed-shift cancellation, parity advance or prime-pair theorem is claimed.

1 Scope and the native entrance

Fix a nonzero admissible shift h_0 , a smooth compactly supported weight W , and $0 < \delta < 1/2$. The TPC-15 hard packet has the opened form

$$B_{h_0, \delta}(X) = \sum_{\alpha=(\ell, k, d) \in \mathcal{A}_X} c_X(\alpha), \quad c_X(\alpha) = -\Lambda(\ell) \mu(d) r_Q(\ell k + h_0) W(\ell k / X), \quad (1)$$

where $\ell > U$, $k > V$, $d \mid k$, and $d \leq V$ [5, 9]. The divisor d is part of the key: compressing (ℓ, k, d) to (ℓ, k) preserves the scalar in (1) but loses the provenance needed by later content and prefix operations.

TPC-119 encodes a complete later analysis by a matrix M_X whose columns are the atoms of $\mathcal{A}_X$ and whose rows are final leaves. Its sufficient conservation condition is

$$\mathbf{1}^T M_X = \mathbf{1}^T. \quad (2)$$

The remaining difficulty is not linear algebra: it is the absence of one object proving that every intermediate branch has been included with the same h_0 , normalization and native source. The present paper replaces a monolithic final archive by a composable path archive. It does not estimate any coefficient in (1).

Definition 1.1 (Claim levels). An exact finite identity independent of the arithmetic meaning of the labels is L0. A verified identification of those labels with the literal TPC packet is L1. A growing, coefficient-specific, fixed- h_0 arithmetic power saving is L2.

2 Typed stages and the path-expanded archive

Let

$$R_0 = \mathcal{A}_X, \quad R_1, \dots, R_s = \mathcal{L}_X$$

be finite record sets. A record may contain mathematical data in addition to its identifier. For each j , let

$$T_j : \mathbb{C}^{R_{j-1}} \longrightarrow \mathbb{C}^{R_j}$$

be the exact sparse matrix of the j -th declared operation. Rows are outgoing records and columns are incoming records. Thus

$$M = T_s T_{s-1} \cdots T_1$$

is the aggregate leaf matrix.

Definition 2.1 (Path-expanded archive). A path is a tuple

$$p = (r_0, r_1, \dots, r_s), \quad r_j \in R_j,$$

such that $T_j(r_j, r_{j-1}) \neq 0$ for every j . Its multiplier is

$$m(p) = \prod_{j=1}^s T_j(r_j, r_{j-1}).$$

The path-expanded archive stores the tuple, its multiplier, its native source r_0 , and the typed metadata on every edge. Paths remain distinct even when they end at the same leaf.

Proposition 2.2 (Path expansion). *For every $r_s \in R_s$ and $r_0 \in R_0$,*

$$M(r_s, r_0) = \sum_{\substack{p: p_0=r_0 \\ p_s=r_s}} m(p).$$

Proof. This is the entrywise definition of the product $T_s \cdots T_1$, with all intermediate indices displayed. $\square$

The path form is deliberately more redundant than M . It records whether an entry arose from the declared branches, whereas a final matrix entry records only their sum. This distinction is essential when opposite errors cancel.

3 Local conservation implies exact global reconnection

Write $\mathbf{1}_j$ for the all-ones column vector on R_j .

Definition 3.1 (Stage conservation). Stage j is coefficientwise conservative when

$$\mathbf{1}_j^T T_j = \mathbf{1}_{j-1}^T. \tag{3}$$

Equivalently, the outgoing multipliers above every incoming record sum exactly to one.

Theorem 3.2 (Local-to-global path conservation). *If (3) holds for every $1 \leq j \leq s$, then*

$$\boxed{\mathbf{1}_s^T M = \mathbf{1}_0^T.}$$

Consequently, for every coefficient vector $c \in \mathbb{C}^{R_0}$,

$$\mathbf{1}_s^T M c = \mathbf{1}_0^T c.$$

In particular, taking $c = c_X$ from (1) gives exact scalar reconnection to $B_{h_0, \delta}(X)$.

Proof. Repeatedly use (3):

$$\mathbf{1}_s^T T_s \cdots T_1 = \mathbf{1}_{s-1}^T T_{s-1} \cdots T_1 = \cdots = \mathbf{1}_0^T.$$

Multiplying by c proves the scalar identity. No positivity, orthogonality or probabilistic interpretation is used. $\square$

Corollary 3.3 (Retained–soft identity). *Suppose $R_s = R_s^{\text{ret}} \sqcup R_s^{\text{soft}}$ is a disjoint typed partition. Let M^{ret} and M^{soft} be obtained from M by retaining the corresponding rows and setting the other rows to zero. Under theorem 3.2,*

$$B_{h_0, \delta}(X) = \mathbf{1}^T M^{\text{ret}} c_X + \mathbf{1}^T M^{\text{soft}} c_X.$$

Proof. The row partition gives $M = M^{\text{ret}} + M^{\text{soft}}$; now apply theorem 3.2. $\square$

Remark 3.4 (What the theorem does not estimate). The corollary is exact even when both terms are large. To obtain the analytic H8 conclusion, a separate complete theorem must prove $\mathbf{1}^T M^{\text{soft}} c_X = o(X)$ in the original normalization. Local conservation does not provide that bound.

4 The exact defect telescope

Define the stage defect

$$d_j^T = \mathbf{1}_j^T T_j - \mathbf{1}_{j-1}^T.$$

Use the convention $T_0 \cdots T_1 = I_{R_0}$.

Theorem 4.1 (Conservation-defect telescope). *For arbitrary compatible stage matrices,*

$$\boxed{\mathbf{1}_s^T M - \mathbf{1}_0^T = \sum_{j=1}^s d_j^T T_{j-1} \cdots T_1.} \quad (4)$$

Proof. Insert and subtract $\mathbf{1}_{s-1}^T T_{s-1} \cdots T_1$:

$$\mathbf{1}_s^T T_s \cdots T_1 - \mathbf{1}_0^T = d_s^T T_{s-1} \cdots T_1 + \mathbf{1}_{s-1}^T T_{s-1} \cdots T_1 - \mathbf{1}_0^T.$$

Iterating proves (4). $\square$

Corollary 4.2 (First failing conservative stage). *If stages $1, \dots, j-1$ satisfy (3) and $d_j \neq 0$, then the path archive contains an exact local failure certificate at stage j . This stops the declared locally conservative implementation, even if later matrices happen to cancel its contribution in (4).*

The last clause matters: global conservation is not the converse of stagewise conservation. A path audit asks whether the declared proof operations themselves are exact partitions, not merely whether their aggregate happens to preserve one scalar.

5 Metadata intertwining

Coefficient conservation cannot see a wrong shift or a second normalization. Let K be any discrete metadata set and let

$$\kappa_j : R_j \longrightarrow K$$

record one field. For $q \in K$, let $P_{j,q}$ be the coordinate projection onto records with $\kappa_j = q$.

Definition 5.1 (Metadata intertwining). Stage T_j preserves the field κ when

$$P_{j,q}T_j = T_jP_{j-1,q} \quad (q \in K), \quad (5)$$

where the second occurrence of T_j is understood with the compatible source and target projections. Entrywise, every nonzero edge has $\kappa_j(r_j) = \kappa_{j-1}(r_{j-1})$.

Theorem 5.2 (Composite metadata preservation). *If (5) holds at every stage, then*

$$P_{s,q}M = MP_{0,q} \quad (q \in K).$$

Proof. Move the projection successively through $T_s, \dots, T_1$. □

The fields required by the present route include:

- F1.** the native atom (ℓ, k, d) ;
- F2.** the prescribed shift h_0 and the single inherited physical normalization;
- F3.** every partition, polarization, mask, content, boundary and collision tag;
- F4.** the retained or soft terminal type;
- F5.** for the TPC-121 interface, the raw atom t , determinant bin $\nu(t)$, complex coefficient u_t , and the predeclared X -indexed bin/good-set rule [7];
- F6.** for the TPC-122 interface, the canonical fiber order, arithmetic sign σ , outer weight w , and the content remainder allocation [8].

The last two items are data fields, not exponent theorems. Recording them does not prove either λ_D^{cert} or η_Z^{cert} .

6 Sharp counterexamples

Proposition 6.1 (Scalar coincidence is insufficient). *There are M and c such that $\mathbf{1}^T M c = \mathbf{1}^T c$ but $\mathbf{1}^T M \neq \mathbf{1}^T$.*

Proof. Take

$$M = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}, \quad c = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Both scalar sums equal 2, whereas the column-sum row is $(2, 0)$. □

Proposition 6.2 (Local defects may cancel). *There are nonconservative stages whose composite is the identity.*

Proof. Take

$$T_1 = \begin{pmatrix} 2 & 0 \\ 0 & 1/2 \end{pmatrix}, \quad T_2 = \begin{pmatrix} 1/2 & 0 \\ 0 & 2 \end{pmatrix}.$$

Neither stage has column sums $(1, 1)$, but $T_2 T_1 = I$. □

Proposition 6.3 (Conservation does not imply provenance). *A permutation matrix can satisfy coefficientwise conservation while exchanging two incompatible metadata tags.*

Proof. Every permutation matrix has unit column sums. Permute a record tagged h_0 with one tagged $h_1 \neq h_0$. The coefficient sum is unchanged, but (5) fails for the shift projection. $\square$

These examples rule out three common shortcuts: testing only the actual coefficient vector, testing only the final composite, and testing only untagged weights.

7 Invocation, data requirements and route verdicts

Definition 7.1 (Paper-specific verdicts). GO-INTERFACE means that the complete symbolic family of actual stage tables, not merely finitely many numerical samples, passes [theorems 3.2](#) and [5.2](#). STOP-DECLARED-ROUTE means that an exact defect disproves the declared archive implementation. It is not a global impossibility theorem. NOT-TESTABLE means that a stage table, boundary rule, metadata dictionary or typed comparison object is absent.

Input	Required quantifier	Failure status
Native entrance	Every active $(\ell, k, d) \in \mathcal{A}_X$	Missing divisor provenance is NOT-TESTABLE.
Stage tables	Every declared branch for the symbolic X -family	A missing table is NOT-TESTABLE; an explicit wrong entry is STOP-DECLARED-ROUTE.
Boundary/collision rules	All endpoint and overlap cases	Testing generic interior samples only is NOT-TESTABLE.
Conservation	Entrywise exact arithmetic	A nonzero stage defect is STOP-DECLARED-ROUTE for that implementation.
Metadata	Every nonzero edge	Wrong h_0 , source or normalization is STOP-DECLARED-ROUTE.
Soft estimate	Complete outer family as $X \rightarrow \infty$	Absent analytic estimate leaves H8 open.

The local dictionaries of TPC-105 and TPC-107 explain how provenance and signed filtering should be retained, while TPC-114 and TPC-117 specify later physical grouping and cover objects [1–4]. They do not, by themselves, provide every T_j . Consequently the current repository still lacks the complete growing input needed to run the present audit:

$$\boxed{\text{Archive}_{123} = \text{NOT-TESTABLE from current artifacts.}}$$

This is a dated administrative statement, not a theorem that no alternative exact H8 proof can exist.

8 Reproducible finite audit and claim ledger

The standard-library script `experiments/tpc123_path_archive_audit.py` uses exact rational arithmetic. It checks two conservative stages, their composite, a retained–soft split, and [theorem 4.1](#). It also constructs the three counterexamples above and rejects a wrong shift tag. Its small model is a regression test for the finite theorem; it is not the missing TPC archive.

Statement	Status
Path expansion and local-to-global conservation	Proved exactly, L0.
Defect telescope and counterexamples	Proved exactly, L0.
Required literal TPC metadata	Specified as an L1 interface.
Complete symbolic growing stage archive	Not present; NOT-TESTABLE.
Complete soft $o(X)$ estimate	Not proved.
Fixed- h_0 arithmetic power saving	Not proved; no L2 event.

Explicit exclusions. This paper proves no determinant-energy exponent, no zero-mode exponent, no complete physical frame, no fixed-shift localization bound, no signed affine Möbius cancellation, no endpoint below $1/400$, no parity breakthrough, no Hardy–Littlewood asymptotic, and no prime-pair or twin-prime theorem. TPC-121 and TPC-122 remain conditional exponent interfaces [6–8].

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